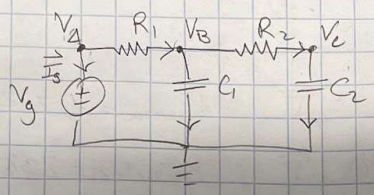
Problem 1



$\vec{V}_A = \vec{V}_g$

KCL

1) $\cancel{+\vec{I}_g} + \vec{I}_{R_1} = 0$

2) $+\vec{I}_{C_1} + \vec{I}_{R_2} - \vec{I}_{R_1} = 0$

3) $-\vec{I}_{R_2} + \vec{I}_{C_2} = 0$

$\vec{I}_g = ?$

$\vec{I}_{R_1} = (\vec{V}_A - \vec{V}_B)/R_1$

$\vec{I}_{C_1} = V_B\, j\omega C_1$

$\vec{I}_{C_2} = V_C\, j\omega C_2$

$\vec{I}_{R_2} = (V_B - V_C)/R_2$

New Equations

1) $\vec{V}_A = \vec{V}_g$

2) $\vec{V}_B\, j\omega C_1 + (V_B - V_C)/R_2 - (\vec{V}_A - \vec{V}_B)/R_1 = 0$

3) $V_C\, j\omega C_2 - (V_B - V_C)/R_2 = 0$

$\Rightarrow$ 2) $\vec{V}_A\left(-\frac{1}{R_1}\right) + \vec{V}_B\left(j\omega C_1 - \frac{1}{R_2} + \frac{1}{R_1}\right) + \vec{V}_C\left(-\frac{1}{R_2}\right) = 0$

3) $\vec{V}_B\left(-\frac{1}{R_2}\right) + \vec{V}_C\left(j\omega C_2 + \frac{1}{R_2}\right) = 0 \Rightarrow \vec{V}_B = \vec{V}_C\left(j\omega C_2 R_2 + 1\right)$

$$\Rightarrow \vec{V}_g\left(\frac{1}{R_1}\right) = \vec{V}_C\left((j\omega C_2 R_2 + 1)\left(j\omega C_1 - \frac{1}{R_2} + \frac{1}{R_1}\right) - \frac{1}{R_2}\right)$$

$$\Rightarrow H(j\omega) = \frac{\vec{V}_{C_2}}{\vec{V}_g} = \frac{\vec{V}_C}{\vec{V}_g} = \frac{1}{R_1\left((j\omega C_2 R_2 + 1)\left(j\omega C_1 - \frac{1}{R_2} + \frac{1}{R_1}\right) - \frac{1}{R_2}\right)}$$

$$\Rightarrow H(j\omega) = \frac{1}{1 + j\omega C_1 R_1 + j\omega C_2 R_2 - \omega^2 C_1 R_1 C_2 R_2 + j\omega R_1 C_2}$$

$$\frac{1}{\left(1 + \frac{j\omega}{a}\right)\left(1 + \frac{j\omega}{b}\right)} = \frac{1}{1 + j\omega\left(\frac{1}{a} + \frac{1}{b}\right) - \omega^2\left(\frac{1}{ab}\right)}$$

$$= \frac{1}{1 + j\omega\left(C_1 R_1 + C_2 R_2 + R_1 C_2\right) - \omega^2\left(C_1 R_1 C_2 R_2\right)}$$

Problem 2

$a = 3000 \qquad b = 20000 \qquad C_1 = C_2 = 1\mu F$

$$\frac{1}{1 + j\omega\left(\frac{1}{a} + \frac{1}{b}\right) - \omega^2\left(\frac{1}{ab}\right)}$$

1) $\frac{1}{a} + \frac{1}{b} = C_1R_1 + C_2R_2 + R_1C_2 = \frac{1}{3000} + \frac{1}{20000} = \frac{23}{6} \times 10^{-4}$

2) $\frac{1}{ab} = C_1R_1C_2R_2 = \frac{5}{3} \times 10^{-8} = C_1^2R_1R_2 = \frac{5}{3} \times 10^{-8}$

1) $C_1(2R_1 + R_2) = \frac{23}{6} \times 10^{-4}$

$\Rightarrow$ 2) $R_1R_2 = \frac{5}{3} \times 10^{4}$

1) $2R_1 + R_2 = \frac{23}{6} \times 10^{2}$

$R_1 = \frac{200}{3} \qquad R_2 = 250$

$R_{1\,actual} = 67.2$

$R_{2\,actual} = 238.35$

Apendix 2

$\Gamma = [\ ]$!

hold on!

1.2

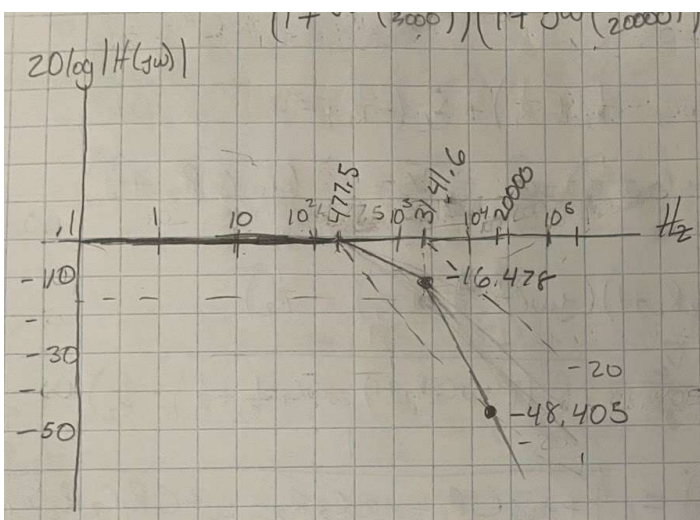
$R_1 \approx 80 \approx 85.458$

$R_2 = 305.68$

$$\frac{1}{\left(1 + j\omega\left(\frac{1}{3000}\right)\right)\left(1 + j\omega\left(\frac{1}{20000}\right)\right)}$$

237

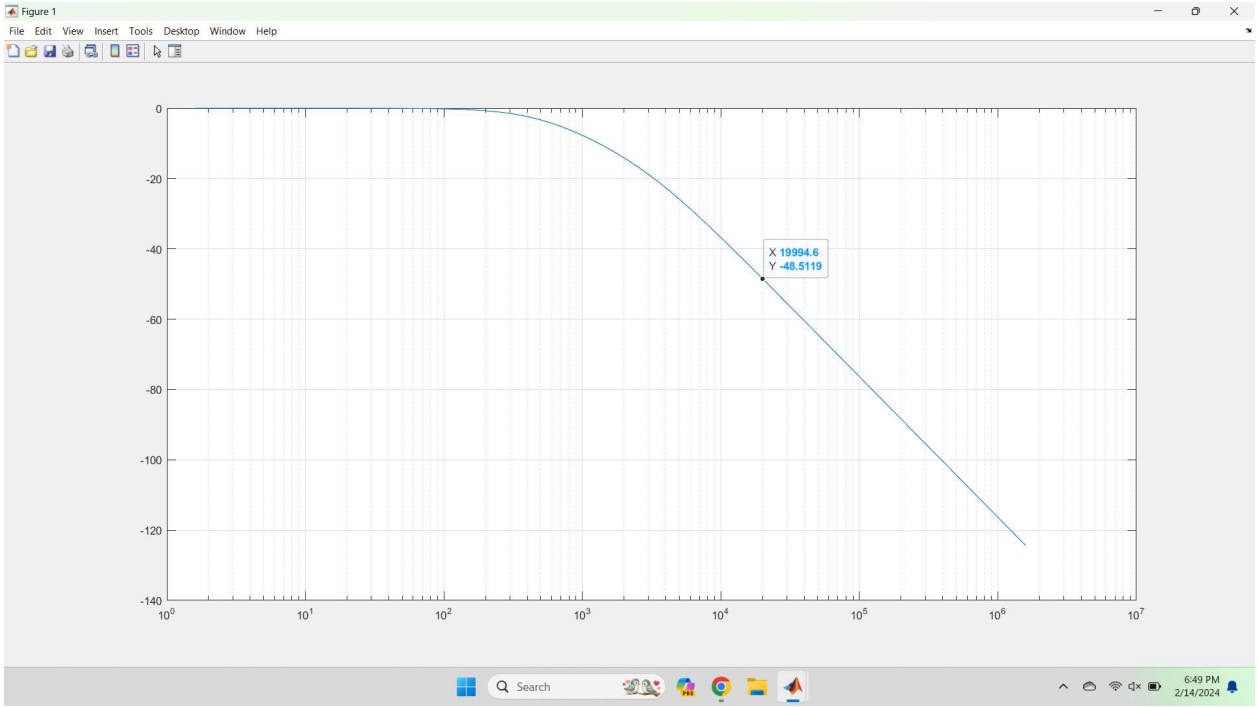
Problem 3



$0 - 20\log \frac{20000/2\pi}{\frac{3000}{2\pi}} = -16.478$

$-16.478 - 40\log \frac{20000}{\frac{20000}{2\pi}}$

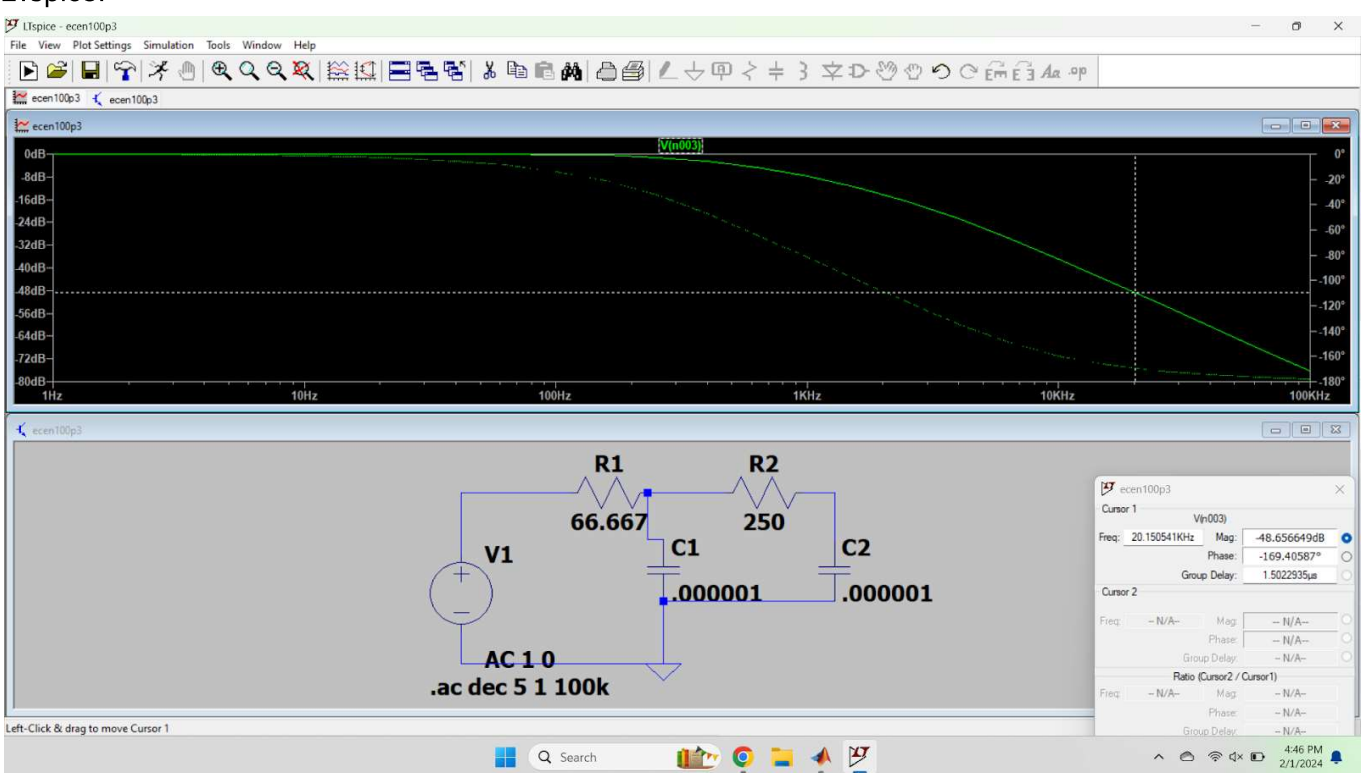
$= -48.405$



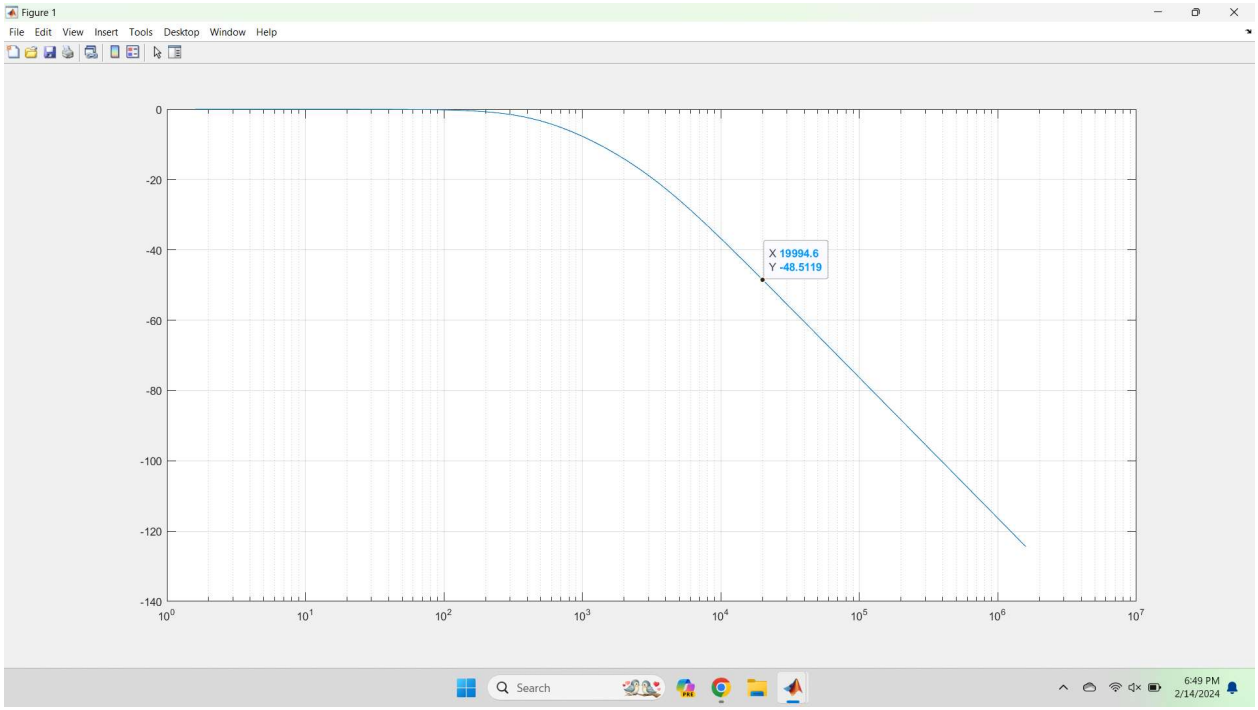
At frequency 20 kHz, the bode plot is calculated for -48.405 dB. The figure above shows that at ~20 kHz it has -48.5119 dB. These are close calculations and can be comparable.

Problem 4

LTspice:

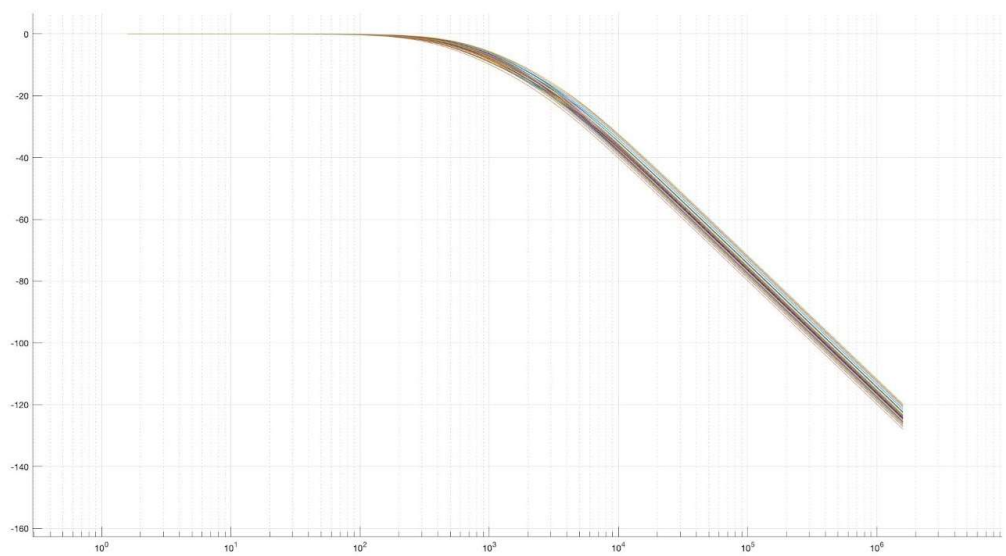


MATLAB:



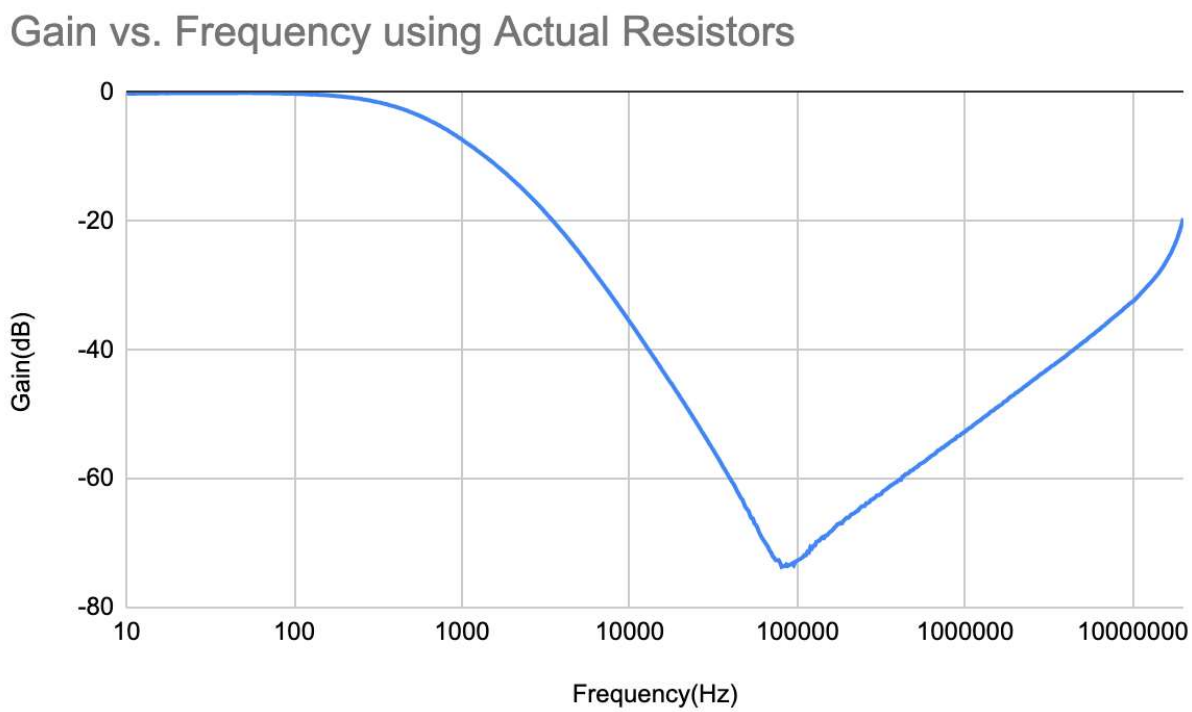
Using the same frequency of 20kHz in LTSpice as in the previous comparison the gain was -48.6 which is very close to the -48.49 from the matlab simulation

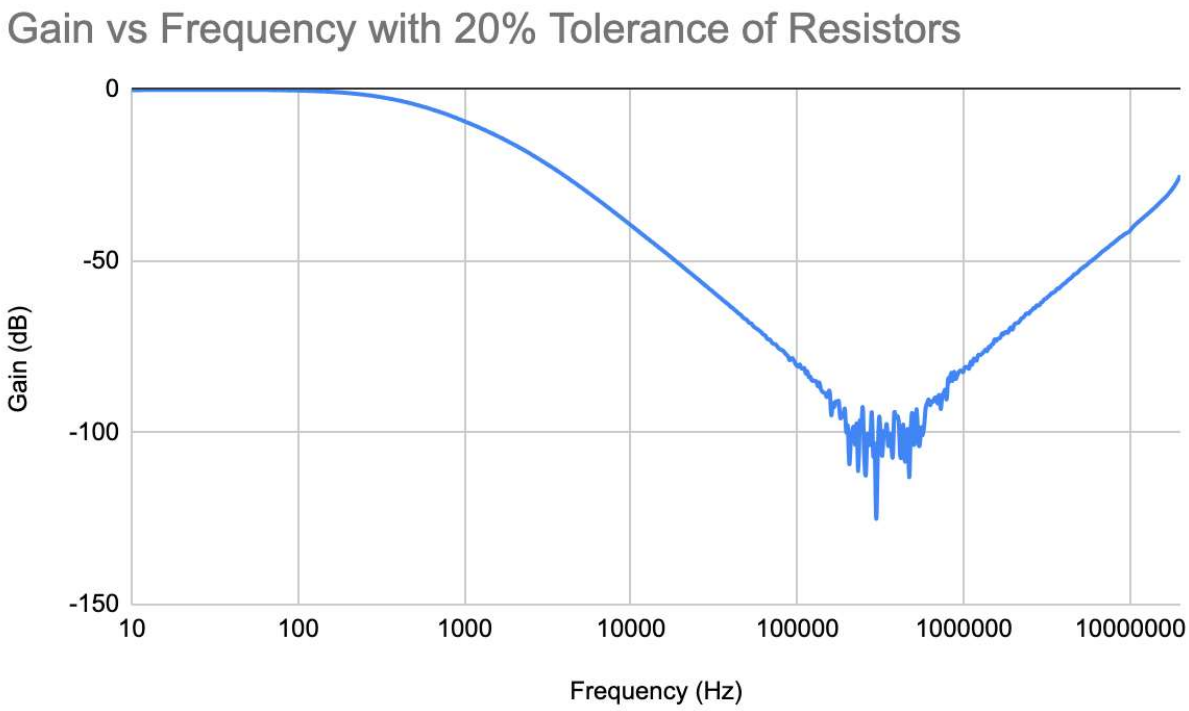
Problem 5



Problem 6

At 20kHz using the actual resistors of R1=67.2Ohms and R2=238.35Ohms the gain was -47.42. This is a reasonable since the calculated resistance should be R1= 66.67Ohms and R2=250Ohms. Using resistors that are 20% more R1=85.458Ohms and R2=305.68Ohms the gain was -51.33. Comparing these values to the matlab script from problem 5 where at 20kHz the gain was -51.99 with the largest resistance and -44.47 was the gain with the lowest resistance.




Gain vs Frequency with 20% Tolerance of Resistors
0
-50
-100
-150
Gain (dB)
10
100
1000
10000
100000
1000000
10000000
Frequency (Hz)